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PROXIMAL CONVOLUTED TUBULE

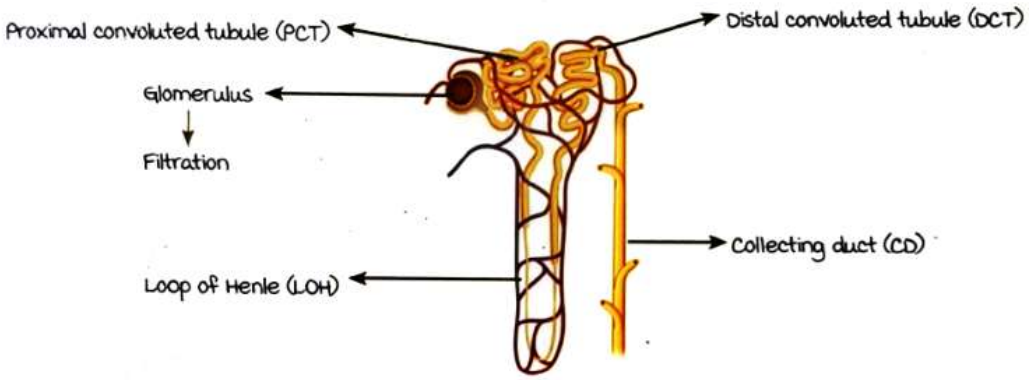
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Structure & Physiology

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Nephron is the functional unit of kidney.

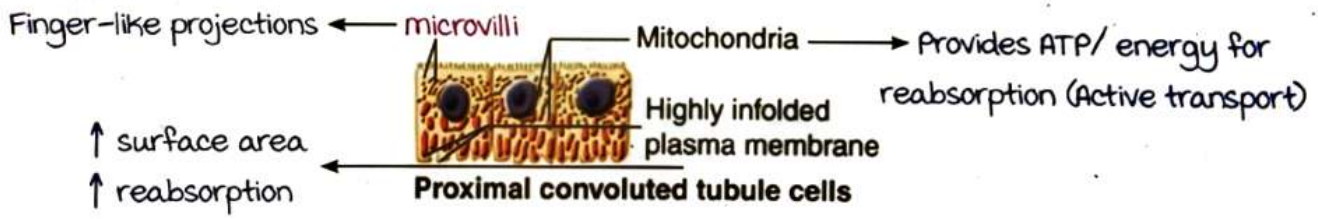
PARTS OF NEPHRON :



Basic renal processes :

Filtration → Reabsorption → Secretion → Excretion {Filtration – (Reabsorption + secretion)}

STRUCTURE OF PROXIMAL TUBULE :



PCT has leaky tight junctions.

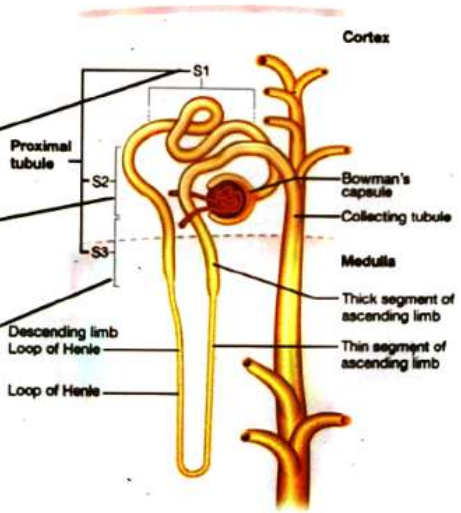
SEGMENTS OF PROXIMAL TUBULE :

S1 : 1st half of proximal convoluted tubule

S2 : 2nd half of PCT + 1st half of proximal straight tubule

S3 : 2nd half of proximal straight tubule

more prone to ischemia (Eg : Shock)



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Reabsorption

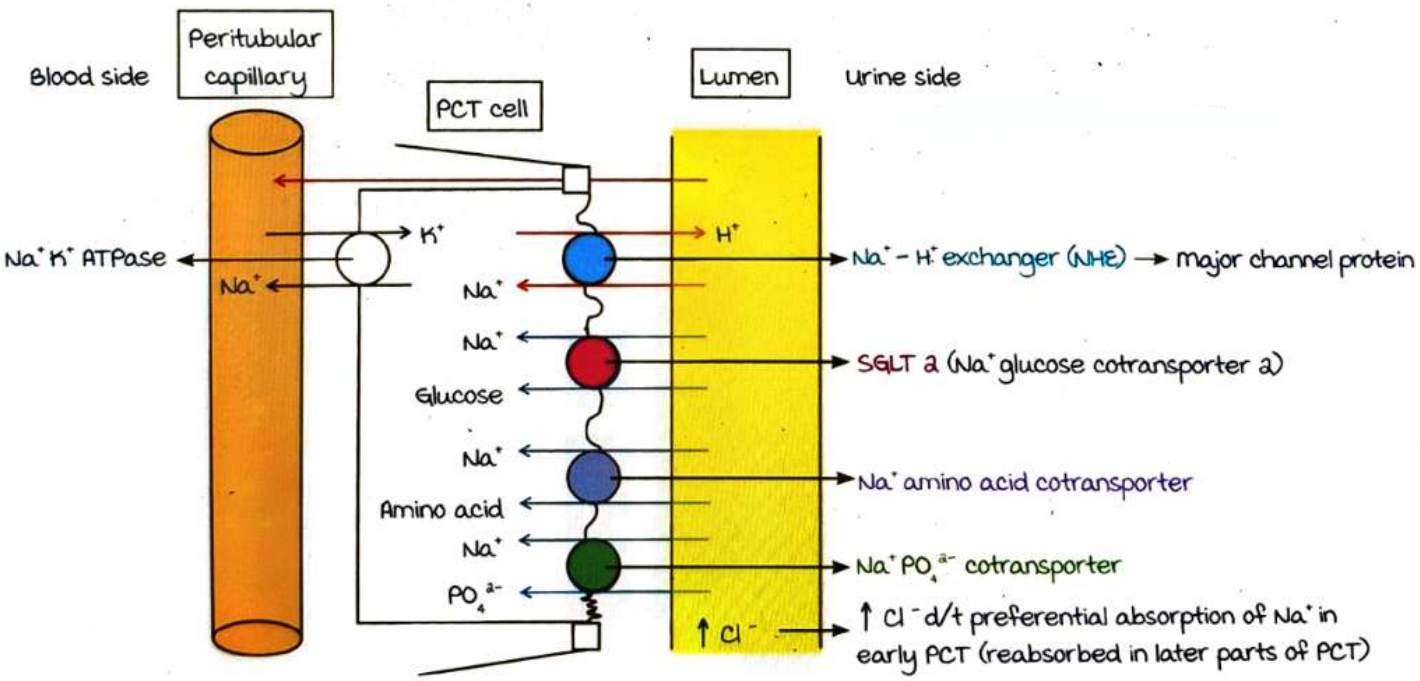
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Solute/Substance	Proportion of filtered substance reabsorbed
NaCl	67 % (2/3rd)
H_2O	67 % (2/3rd)
K+	67 % (2/3rd)
Urea	67 % (2/3rd)
Ca^{2+}	70 %
Phosphate & bicarbonate	80 %
Glucose, Amino acids	100 %

Equal amounts of Na^+ and H_2O reabsorbed : PCT fluid is isotonic.

Note : PCT is the only segment absorbing glucose.

SODIUM REABSORPTION



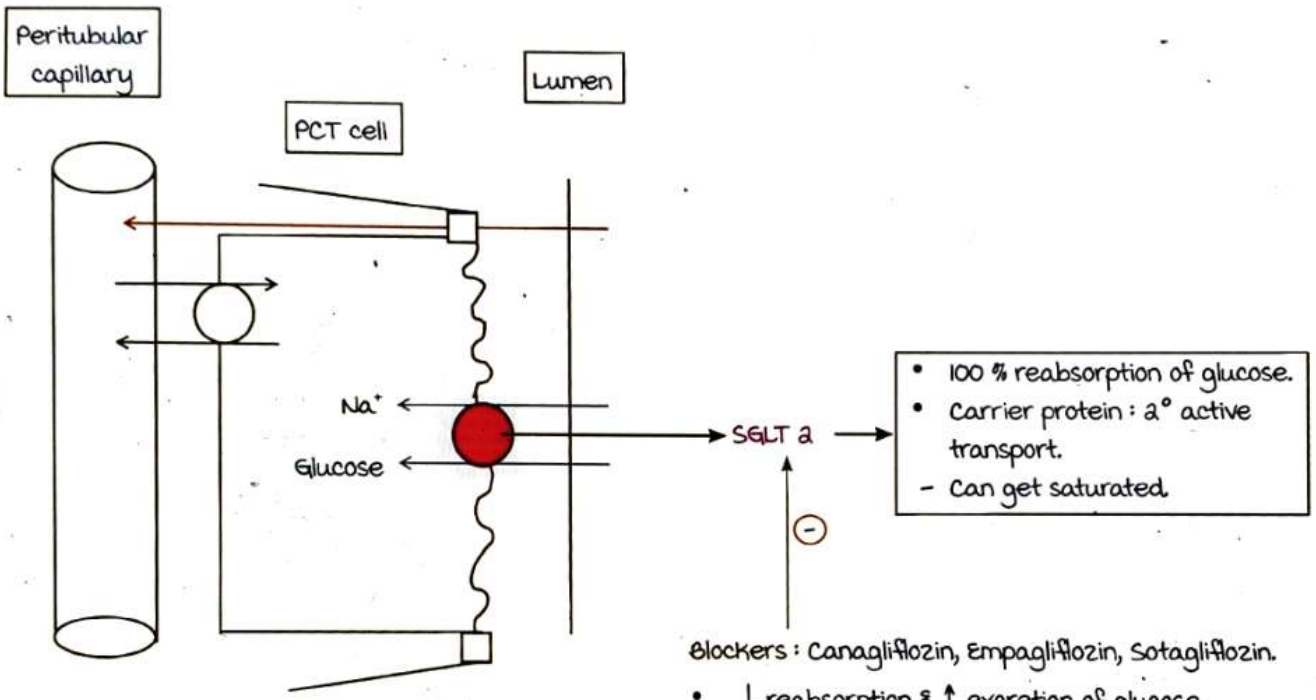
Note : SGLT-1 is found in GIT.

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GLUCOSE REABSORPTION



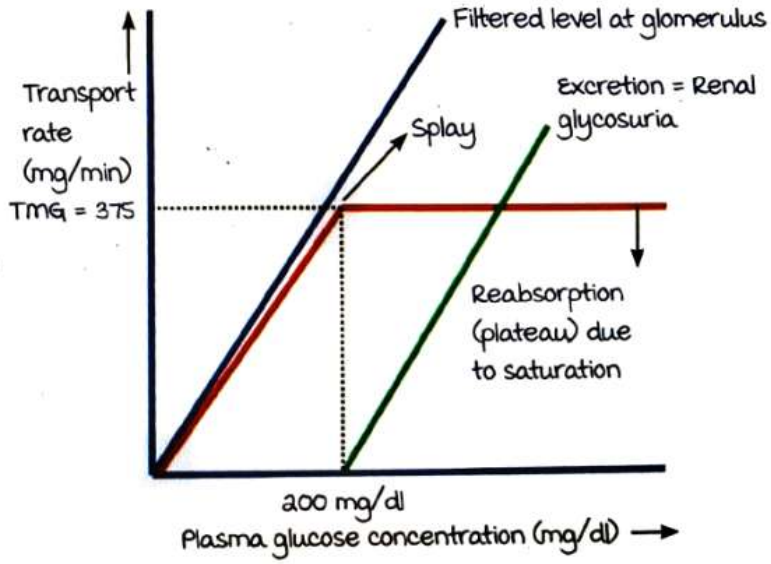
Saturation of SGLT-2 :

- At 200 mg/dL of plasma glucose → 375 mg/dl of glucose is transported (Transport maximum)

Saturation of SGLT-2

Excretion of excess glucose

- Splay : Deviation in reabsorption curve d/t diverse/heterogeneous nephrons → Different TmG

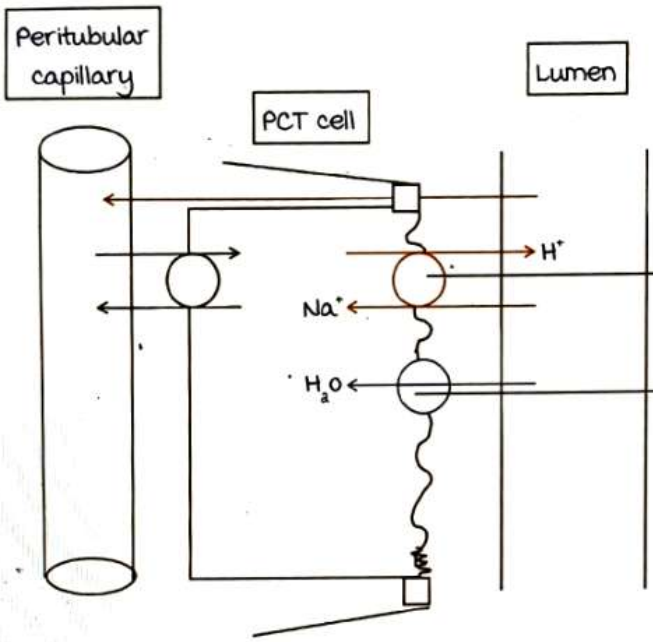


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WATER REABSORPTION



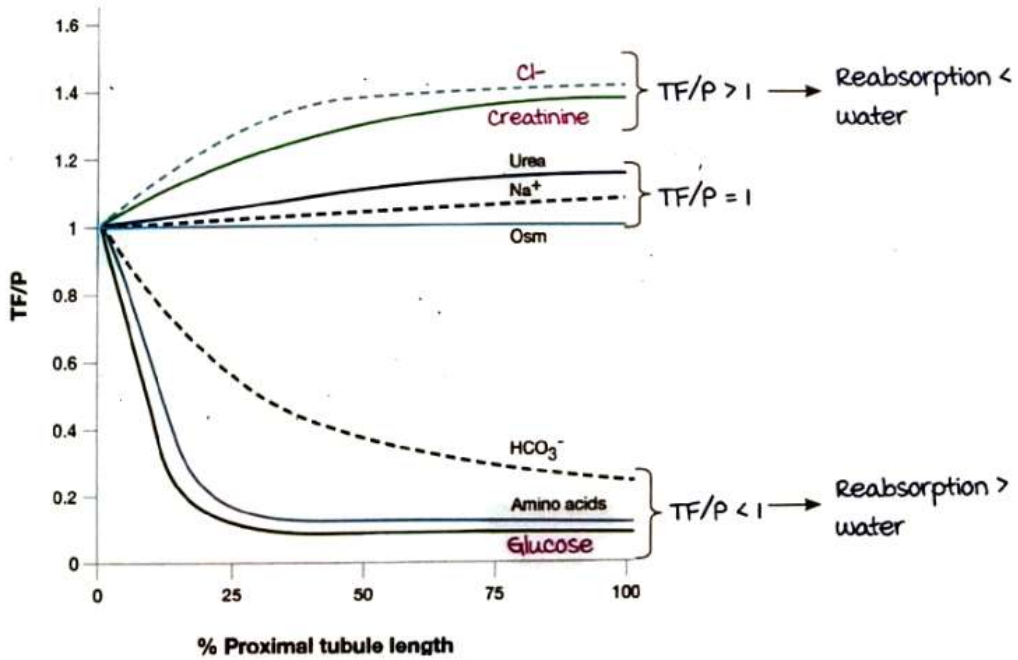
Obligatory water reabsorption : Osmosis.
- Na^+ follows H_2O.
- Not influenced by hormones.

Note : Facultative water reabsorption occurs in collecting duct (Facilitated diffusion).
- Hormone dependent : ADH.

TUBULAR FLUID PLASMA RATIO (TF/P)

- ↑ reabsorption ← ↑ plasma conc. (P), ↓ tubular conc. (TF)
- TF/P = 1 or close to 1

  Reabsorption is same as that of water in PCT.



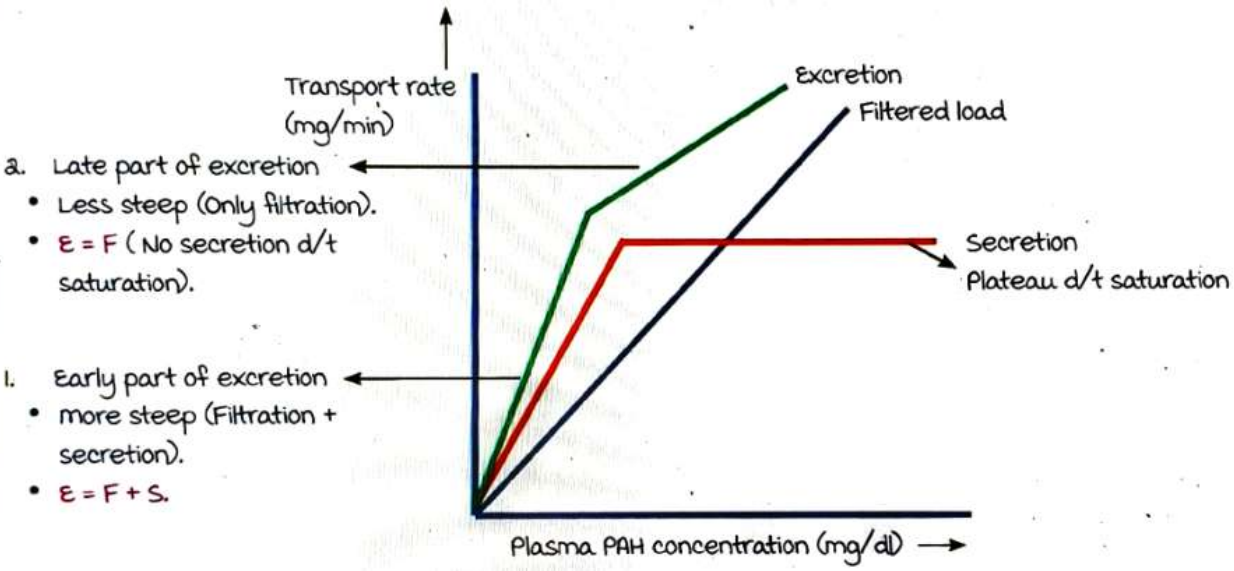
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Secretion

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- PCT secretes para amino hippuric acid (PAH) via carrier protein → Saturation seen.
- Excretion of PAH (E) = Filtered load (F) + secretion (S).



2. Late part of excretion
   - Less steep (Only filtration).
   - E = F (No secretion d/t saturation).

1. Early part of excretion
   - more steep (Filtration + secretion).
   - E = F + S.

Feedback